

Customer Churn Analysis in European Banking

Machine Learning Internship Project

Abstract

This study analyzes customer churn behavior in a European retail bank using segmentation-based analytics. The analysis identifies churn patterns across geography, age groups, tenure, credit score, and engagement indicators to support retention strategy planning.

Introduction

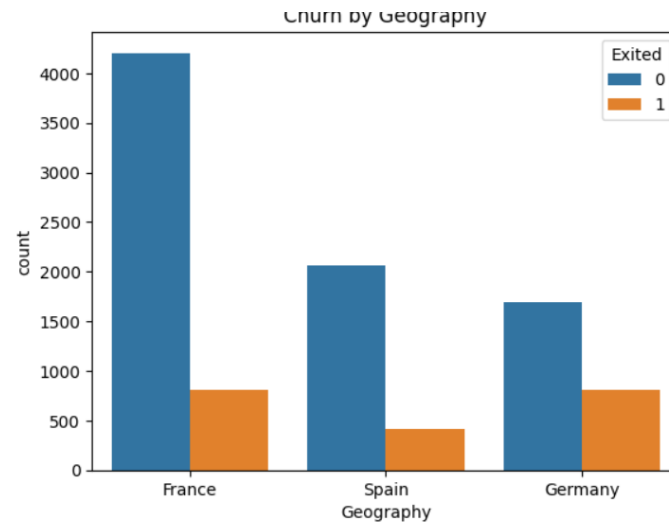
Customer churn leads to revenue loss and increased acquisition costs. Banks require targeted retention strategies based on customer behavior rather than generic marketing approaches.

Methodology

The dataset was cleaned and segmented into age groups, tenure categories, credit score bands, and balance levels. Churn rates were calculated using grouped statistical analysis and visualizations.

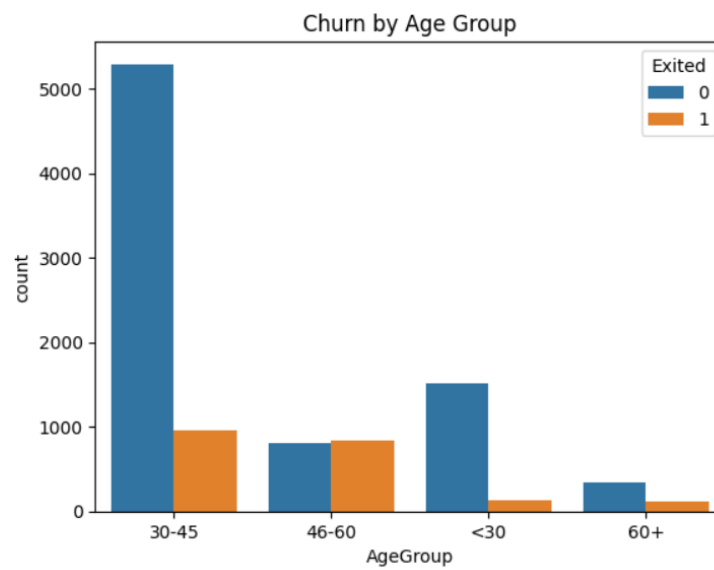
Results

Churn by Geography



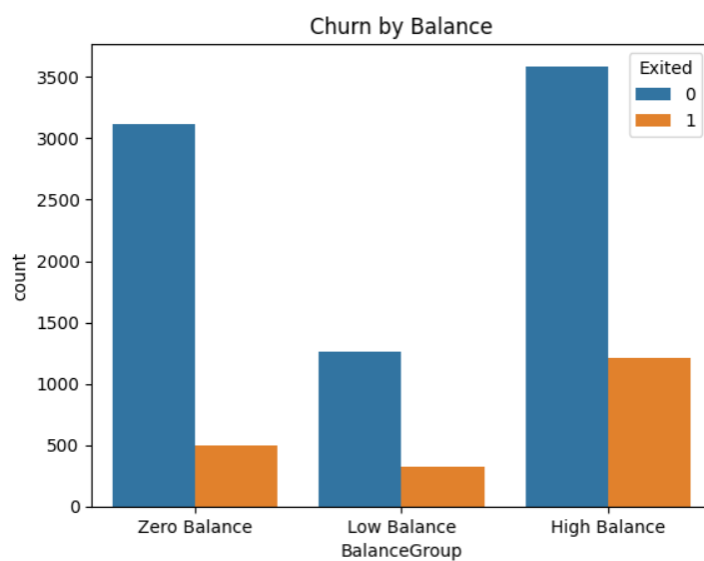
Germany shows higher churn compared to France and Spain.

Churn by Age Group



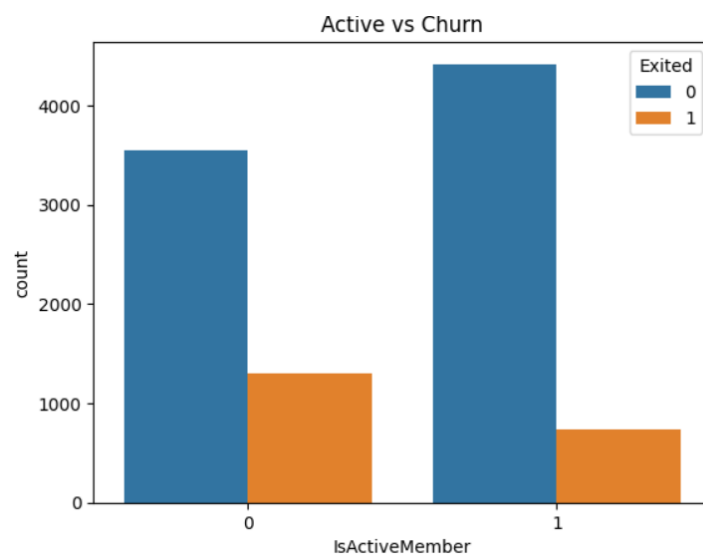
Older customers show increased probability of exiting the bank.

Churn by Balance



Customers with zero or low balance exhibit higher churn behavior.

Active vs Inactive Members



Inactive members are significantly more likely to churn.

Conclusion

Segmentation-driven churn analysis enables banks to identify high-risk customers and design targeted engagement programs.